

Patient Name:	MR. BIRENDRA KR TRIPATHI	Sample ID:	2600132470
Age / Gender:	49 Years / Male	Lab ID:	KOL2604200324
Ref. Doctor:	N/A	Report Date:	20-May-2026 06:11 AM

NGS HEREDITARY CANCER PANEL ON THE ILLUMINA NOVASEQ X NGS PLATFORM

CLINICAL INDICATION & SAMPLE DESCRIPTION
Clinical Indication: Gene & Transcript Location Variant Zygosity/ Inheritance Phenotype Clinical Significance
Sample Description: EDTA PERIPHERAL BLOOD

KEY GENETIC FINDINGS

Gene & Transcript	Location	Variant	Zygosity	Phenotype	Clinical Significance
MSH2 (+) NM 000251.3 _	2p21 (? 47,410,089- _ 47,416,441 ?) _ Exon 3-6	c.367-? c.1076+? _ 6.4kb	Heterozygous /Autosomal Dominant	Lynch syndrome 1	Pathogenic
HBB (-) NM 000518.5 _ (chr11:g.5227071T>G)	Exon 1	c.-50A>C (5' prime UTR variant) 49x/51%	Heterozygous /Autosomal Recessive	c.-50A>C (5' prime UTR variant) 49x/51%	Pathogenic (PM2, PM3, PP5)
Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).	Variant			Variant
Benign	A variant which is known not to be responsible for disease has been detected. Generally, no further action is warranted on such variants when detected.	Benign			Benign
Likely Benign	A variant which is very unlikely to contribute to the development of disease however, the	Likely Benign			Likely Benign

Gene & Transcript	Location	Variant	Zygosity	Phenotype	Clinical Significance
	scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of Pathogenicity.				
Pathogenic	A disease-causing variation in a gene which can explain the patient's symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.	Pathogenic			Pathogenic
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.	Likely Pathogenic			Likely Pathogenic
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific	Variant of Uncertain Significance			Variant of Uncertain Significance

Gene & Transcript	Location	Variant	Zygosity	Phenotype	Clinical Significance
	evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.				
MT-ATP6	100.00%	MT-ND6	100.00%	MT-TL1	100.00%
MT-ATP8	100.00%	MT-RNR1	100.00%	MT-TL2	100.00%
MT-CO1	100.00%	MT-RNR2	100.00%	MT-TM	100.00%
MT-CO2	100.00%	MT-TA	100.00%	MT-TN	100.00%
MT-CO3	100.00%	MT-TC	100.00%	MT-TP	100.00%
MT-CYB	100.00%	MT-TD	100.00%	MT-TQ	100.00%
MT-ND1	100.00%	MT-TE	100.00%	MT-TR	100.00%
MT-ND2	100.00%	MT-TF	100.00%	MT-TS1	100.00%
MT-ND3	100.00%	MT-TG	100.00%	MT-TS2	100.00%
MT-ND4	100.00%	MT-TH	100.00%	MT-TT	100.00%
MT-ND4L	100.00%	MT-TI	100.00%	MT-TV	100.00%
MT-ND5	100.00%	MT-TK	100.00%	MT-TW	100.00%
MT-TY	100.00%				
RAD50	100.00%	CDK4	100.00%	CDKN1B	100.00%
CDKN2A	100.00%	CHEK2	100.00%	FLCN	100.00%
FH	100.00%	MSH6	100.00%	APC	100.00%
EPCAM	100.00%	SMAD4	100.00%	MAX	100.00%
MEN1	100.00%	MET	100.00%	MITF	100.00%
MLH1	100.00%	MRE11	100.00%	MSH2	100.00%
MUTYH	100.00%	NBN	100.00%	ATM	100.00%
NF1	100.00%	PMS2	100.00%	SDHAF2	100.00%
TMEM127	100.00%	PRKAR1A	100.00%	PTCH1	100.00%
PTEN	100.00%	BARD1	100.00%	RAD51C	100.00%

Gene & Transcript	Location	Variant	Zygosity	Phenotype	Clinical Significance
RAD51D	100.00%	RB1	100.00%	RET	100.00%
SDHB	100.00%	BLM	100.00%	BMPR1A	100.00%
BRCA1	100.00%	BRCA2	100.00%	STK11	100.00%
TP53	100.00%	TSC1	100.00%	TSC2	100.00%
VHL	100.00%	XRCC2	100.00%	CDC73	100.00%
PALB2	100.00%	BAP1	100.00%	BRIP1	100.00%
CDH1	100.00%	AXIN2	100.00%	CTNNA1	100.00%
DICER1	100.00%	GREM1	100.00%	HOXB13	100.00%
KIT	100.00%	MSH3	100.00%	NTHL1	100.00%
PDGFRA	100.00%	POLD1	100.00%	POLE	100.00%
SDHA	100.00%	SDHC	100.00%	SDHD	100.00%
SMARCA4	100.00%	VPS13B	100.00%	RNASEL	100.00%
ELAC2	100.00%	MSR1	100.00%	PRSS1	100.00%
SPINK1	100.00%	ABRAXAS1	100.00%	ACVRL1	100.00%
AKT1	100.00%	BUB1B	100.00%	CASR	100.00%
CDKN1C	100.00%	CEBPA	100.00%	DDX41	100.00%
DIS3L2	100.00%	EGFR	100.00%	ETV6	100.00%
EXT1	100.00%	EXT2	100.00%	FANCC	100.00%
GALNT12	100.00%	GATA2	100.00%	GPC3	100.00%
HNF1A	100.00%	HNF1B	100.00%	HRAS	100.00%
KIF1B	100.00%	MC1R	100.00%	MLH3	100.00%
NF2	100.00%	PHOX2B	100.00%	PIK3CA	100.00%
PMS1	100.00%	POT1	100.00%	PTCH2	100.00%
RECQL	100.00%	REST	100.00%	RNF43	100.00%
RPS20	100.00%	RUNX1	100.00%	SAMD9L	100.00%
SMARCA2	100.00%	SMARCB1	100.00%	SMARCE1	100.00%
SUFU	100.00%	TERT Pag	100.00% e 6 of 8	TGFBR2	100.00%

Gene & Transcript	Location	Variant	Zygosity/ Inheritance	Phenotype	Clinical Significance
MSH2 (+) NM 000251.3 _	2p21 (? 47,410,089- _ 47,416,441 ?) _ Exon 3-6	c.367-? c.1076+? _ 6.4kb	Heterozygous /Autosomal Dominant	Lynch syndrome 1	Pathogenic

Dr.	Parag	Tamhankar
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Gene & Transcript	Location	Variant & Depth/ VAF	Zygosity/ Inheritance	Phenotype	Clinical Significance
HBB (-) NM 000518.5 _ (chr11:g.5227071T>G)	Exon 1	c.-50A>C (5' prime UTR variant) 49x/51%	Heterozygous /Autosomal Recessive	Beta-thalassemia HBB	Pathogenic (PM2, PM3, PP5)

Dr.Nirmal	A.	Vaniawala
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Dr.	Parag	Tamhankar
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Dr.Nirmal	A.	Vaniawala
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Dr.	Parag	Tamhankar
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*Variant classification as per ACMG guidelines:	
Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
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Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Dr.Nirmal	A.	Vaniawala
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Dr.	Parag	Tamhankar
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Title	Data
Total data generated (Gb)	8.09
Total Reads (%)	100.00
Total reads aligned (%)	99.30
Data >Q30 (%)	96.93

Mean Depth		Percentage target base pairs covered	
80.16x	1x	5x	20x
	98.71%	98.42%	97.85%

Gene	Region covered	Gene	Region covered	Gene	Region covered
MT-ATP6	100.00%	MT-ND6	100.00%	MT-TL1	100.00%
MT-ATP8	100.00%	MT-RNR1	100.00%	MT-TL2	100.00%
MT-CO1	100.00%	MT-RNR2	100.00%	MT-TM	100.00%
MT-CO2	100.00%	MT-TA	100.00%	MT-TN	100.00%
MT-CO3	100.00%	MT-TC	100.00%	MT-TP	100.00%
MT-CYB	100.00%	MT-TD	100.00%	MT-TQ	100.00%
MT-ND1	100.00%	MT-TE	100.00%	MT-TR	100.00%
MT-ND2	100.00%	MT-TF	100.00%	MT-TS1	100.00%
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MT-ND4L	100.00%	MT-TI	100.00%	MT-TV	100.00%
MT-ND5	100.00%	MT-TK	100.00%	MT-TW	100.00%
MT-TY	100.00%				

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Dr.	Parag	Tamhankar
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Gene	Region covered	Gene	Region covered	Gene	Region covered
RAD50	100.00%	CDK4	100.00%	CDKN1B	100.00%
CDKN2A	100.00%	CHEK2	100.00%	FLCN	100.00%
FH	100.00%	MSH6	100.00%	APC	100.00%
EPCAM	100.00%	SMAD4	100.00%	MAX	100.00%
MEN1	100.00%	MET	100.00%	MITF	100.00%
MLH1	100.00%	MRE11	100.00%	MSH2	100.00%
MUTYH	100.00%	NBN	100.00%	ATM	100.00%
NF1	100.00%	PMS2	100.00%	SDHAF2	100.00%
TMEM127	100.00%	PRKAR1A	100.00%	PTCH1	100.00%
PTEN	100.00%	BARD1	100.00%	RAD51C	100.00%
RAD51D	100.00%	RB1	100.00%	RET	100.00%

Gene	Region covered	Gene	Region covered	Gene	Region covered
SDHB	100.00%	BLM	100.00%	BMPR1A	100.00%
BRCA1	100.00%	BRCA2	100.00%	STK11	100.00%
TP53	100.00%	TSC1	100.00%	TSC2	100.00%
VHL	100.00%	XRCC2	100.00%	CDC73	100.00%
PALB2	100.00%	BAP1	100.00%	BRIP1	100.00%
CDH1	100.00%	AXIN2	100.00%	CTNNA1	100.00%
DICER1	100.00%	GREM1	100.00%	HOXB13	100.00%
KIT	100.00%	MSH3	100.00%	NTHL1	100.00%
PDGFRA	100.00%	POLD1	100.00%	POLE	100.00%
SDHA	100.00%	SDHC	100.00%	SDHD	100.00%
SMARCA4	100.00%	VPS13B	100.00%	RNASEL	100.00%
ELAC2	100.00%	MSR1	100.00%	PRSS1	100.00%
SPINK1	100.00%	ABRAXAS1	100.00%	ACVRL1	100.00%
AKT1	100.00%	BUB1B	100.00%	CASR	100.00%
CDKN1C	100.00%	CEBPA	100.00%	DDX41	100.00%
DIS3L2	100.00%	EGFR	100.00%	ETV6	100.00%
EXT1	100.00%	EXT2	100.00%	FANCC	100.00%
GALNT12	100.00%	GATA2	100.00%	GPC3	100.00%
HNF1A	100.00%	HNF1B	100.00%	HRAS	100.00%
KIF1B	100.00%	MC1R	100.00%	MLH3	100.00%
NF2	100.00%	PHOX2B	100.00%	PIK3CA	100.00%
PMS1	100.00%	POT1	100.00%	PTCH2	100.00%
RECQL	100.00%	REST	100.00%	RNF43	100.00%
RPS20	100.00%	RUNX1	100.00%	SAMD9L	100.00%
SMARCA2	100.00%	SMARCB1	100.00%	SMARCE1	100.00%
SUFU	100.00%	TERT Pag	100.00% e 6 of 8	TGFBR2	100.00%

Dr.Nirmal

A.

Vaniawala

Dr.

Para

g Tamhankar

TRIP13	100.00%	WRN	100.00%	WT1	100.00%
XRCC3	100.00%	CDKN2B	100.00%	AIP	100.00%
ALK	100.00%	ANKRD26	100.00%	BRAF	100.00%

TRIP13	100.00%	WRN	100.00%	WT1	100.00%
CBL	100.00%	CD70	100.00%	CEP57	100.00%
CYLD	100.00%	DDB2	100.00%	DKC1	100.00%
EFL1	100.00%	ELANE	100.00%	ERCC1	100.00%
ERCC2	100.00%	ERCC3	100.00%	ERCC4	100.00%
ERCC5	100.00%	EXO1	100.00%	EZH2	100.00%
FAM111B	100.00%	FANCA	100.00%	FANCB	100.00%
FANCD2	100.00%	FANCE	100.00%	FANCF	100.00%
FANCG	100.00%	FANCI	100.00%	FANCL	100.00%
FANCM	100.00%	GPR101	100.00%	HAVCR2	100.00%
IKZF1	100.00%	KITLG	100.00%	KRAS	100.00%
LZTR1	100.00%	MAP2K1	100.00%	MAP2K2	100.00%
NRAS	100.00%	NSD1	100.00%	NSUN2	100.00%
PAX5	100.00%	POLH	100.00%	PPM1D	100.00%
PRF1	100.00%	PTPN11	100.00%	RAF1	100.00%
RASA2	100.00%	RECQL4	100.00%	RHBDF2	100.00%
RIT1	100.00%	RRAS	100.00%	SAMD9	100.00%
SBDS	100.00%	SHOC2	100.00%	SLX4	100.00%
SOS1	100.00%	SOS2	100.00%	SPRED1	100.00%
SRP72	100.00%	TINF2	100.00%	XPA	100.00%
XPC	100.00%	CPA1	100.00%	RFWD3	100.00%
BTNL2	100.00%	NHEJ1	100.00%	RINT1	100.00%
AR	100.00%	CD82	100.00%	ATR	100.00%
PALLD	100.00%	ZFHX3	100.00%	AXIN1	100.00%
TGFB1	100.00%	ENG	100.00%	RAD51	100.00%
LIG4	100.00%	TOX3	100.00%	LSP1	100.00%
RASAL1	100.00%	MXI1	100.00%	CFTR	100.00%
FGFR2	100.00%	MAD2L2	100.00%	MYSM1	100.00%
MAP3K1	100.00%	UBE2T	100.00%	ACD	100.00%
KLHDC8B	100.00%	NPAT	100.00%	TERF2IP	100.00%
ADA	100.00%	CARD11	100.00%	CASP10	100.00%
CD27	100.00%	CTLA4	100.00%	DOCK8	100.00%
FAS	100.00%	FASLG	100.00%	ITK	100.00%
MAGT1	100.00%	PIK3CD	100.00%	SH2D1A	100.00%

TRIP13	100.00%	WRN	100.00%	WT1	100.00%
APCS	100.00%	STXBP2	100.00%	TNFRSF13B	100.00%
WAS	100.00%	CARMIL2	100.00%	CASP8	100.00%
CTPS1	100.00%	FADD	100.00%	FCHO1	100.00%
IL2RA	100.00%	IL2RB	100.00%	MCM4	100.00%

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PIK3R1	100.00%	PRKCD	100.00%	RAC2	100.00%
RASGRP1	100.00%	RHOH	100.00%	STAT3	100.00%
STK4	100.00%	TPP2	100.00%	XIAP	100.00%
CREBBP	100.00%	MYD88	100.00%	STAT6	100.00%
B2M	100.00%	EP300	100.00%	NOTCH1	100.00%
TET2	100.00%	BCL2	100.00%	NOTCH2	100.00%
TNFAIP3	100.00%	FOXO1	100.00%	TNFRSF14	100.00%
GNA13	100.00%	PIM1	100.00%	CD79A	100.00%
KMT2D	100.00%	CD79B	100.00%	MYC	100.00%
SOCS1	100.00%	FAN1	100.00%	CDK12	100.00%
CHEK1	100.00%	CUL2	100.00%	DIS3	100.00%
ERBB2	100.00%	ESR2	100.00%	TDO2	100.00%
GFAP	100.00%	MED12	100.00%	PPP2R2A	100.00%
RAD54L	100.00%	SLC45A2	100.00%	SPOP	100.00%
TYR	100.00%	MBD4	100.00%	HBB	100.00%

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INTERPRETATION & REMARKS

& Clinical correlation:
MSH2:Deletion Exon 3-6;
The submitted sample shows Heterozygous deletion of exon 3-6 of gene MSH2 on chromosome 2. A 6.4kb deletion detected on cytoband 2p21 (?_47,410,089-47,416,441_?). This variant is absent in our internal database.
Based on above evidence, this variant (MSH2:Deletion Exon 3-6) is classified as Pathogenic variant.
MSH2 Gene Phenotype Description:
Lynch syndrome-1 (LYNCH1), also known as hereditary nonpolyposis colorectal cancer (HNPCC1), is caused by heterozygous mutation in the mismatch repair gene MSH2 (609309) on chromosome 2p21-p16. Lynch syndrome (LYNCH), also called hereditary nonpolyposis colorectal cancer, is defined as an autosomal dominant predisposition

to a spectrum of cancers, primarily of the colorectum and endometrium, which exhibit impaired DNA mismatch repair (MMR) activity (summary by Lynch et al., 2015).
Clinical Indication:
Birendra Tripathi diagnosed case of adenocarcinoma of ascending colon with brain lesion.
Pathogenic deletion identified in relation to phenotype
Page 1 of 8

QR Code for
report verification
Case ID
: 2600132470
Sample Type
: EDTA PERIPHERAL BLOOD
Name
: MR. BIRENDRA KR TRIPATHI (KOL2604200324)
Date & Time Collected : 20-Apr-2026 06:44 AM
Sex/Age
: Male/49 Years
Date & Time Received : 22-Apr-2026 01:03 PM
Bill. Loc.
: OncQuest Laboratories Ltd., New Delhi
Date & Time Reported : 20-May-2026 06:11 AM
Ref. By
:
Report Version
: 1

Indication :
MC-7414
Dr.Nirmal A. Vaniawala
Dr. Salil Vaniawala
Dr. Parag Tamhankar
MD (Path. & Bact.)
M.Sc Ph.D.(Human Genetics) Consulting
Geneticist
DM (Medical Genetics),
MD, DNB (Pediatrics)
Senior Genetic Consultant

Additional Findings:

Gene & Transcript

Location

Variant &

Depth/VAF

Zygosity/

Inheritance

Phenotype

Clinical Significance

HBB (-)

NM_000518.5

(chr11:g.5227071T>G)

Exon 1

c.-50A>C

(5' prime UTR

variant)

49x/51%

Heterozygous

/Autosomal

Recessive

Beta-thalassemia HBB

Pathogenic

(PM2, PM3, PP5)

*Genetic test results are reported based on the recommendations of American College of Medical Genetics

HBB Gene Phenotype Description:

Direct sequencing of specific regions of genomic DNA became feasible with the invention of the polymerase chain reaction

(PCR) which permits amplification of specific regions of DNA. Recently, human mitochondrial DNA was amplified and

directly sequenced. Using a thermostable DNA polymerase of *T. aquaticus* (Saiki, R.K. et al., manuscript in preparation) in

the PCR, we have applied a combination of PCR and direct sequence analysis of the amplified product to a human single-

copy gene. We studied the genomic DNA of five patients with beta-thalassaemia whose mutant alleles were uncharacterized, and found two previously undescribed mutations, along with three known alleles. One new allele is a

frameshift at codons 106-107 and the other is an A-C transversion at the cap site (+1) of the beta-globin gene. This latter is

the first natural mutation observed at the cap site and it occurs in a gene which is poorly expressed.

Case Specific Recommendations:

1.

Segregation analysis of the variant by Sanger /Exome sequencing/ MLPA is recommended.

Pathologist

Senior Geneticist

Medical Director